

Mingxian Li

+86 15698416822 | limx253@mail2.sysu.edu.cn

EDUCATION

Sun Yat-sen University

B.Eng. in Microelectronics Science and Engineering

GPA: 3.7/4.0 | Data Structures: 97/100

Sept. 2024 – Present

Shenzhen, China

RESEARCH INTERESTS

Multimodal Reasoning | 3D Spatial Intelligence | AI for Science

RESEARCH EXPERIENCE

SeeChem: 3D Molecular Reasoning in VLMs

Mar. 2026 – Present

Sun Yat-sen University, HCP Lab

Advisors: Xiaodan Liang, Xiao Dong

Contents: Diagnosing and improving how vision-language models connect 3D molecular geometry with chemical rules.

- Developed **SeeChem**, a diagnostic benchmark with **5,065 questions across 320 molecular instances and five tasks**, aligning spatial grounding, local rule application, and integrated reasoning within shared molecular contexts.
- Constructed **SeeChem-Data** with verifiable supervision across the three reasoning stages, using molecular instances excluded from the benchmark.
- Developed **EasySeeChem-VLM**, integrating **Uni-Mol2** geometry representations into **Qwen3-VL-8B** through a gated projector; evaluated the method alongside **nine external VLMs** with task- and stage-specific analysis.

Achievement: SeeChem manuscript **advanced to Phase 2 review at AAAI 2027**.

PROJECT EXPERIENCE

Chemistry Benchmark and Data Synthesis

Apr. 2026 – Present

Sun Yat-sen University–Huawei Collaboration

Advisors: Xiaodan Liang, Xiao Dong

Contents: Benchmark and data synthesis for advanced university and competition chemistry.

- Curated **high-difficulty chemistry problems** from frontier benchmarks published at leading conferences.
- Annotated **high-quality chain-of-thought (CoT) solutions** with explicit, step-by-step chemical reasoning.

Achievement: Contributed chemistry questions and CoT annotations for **Huawei's model training**.

PUBLICATIONS

[1] **SeeChem: A Diagnostic Benchmark for 3D Chemical Reasoning in Vision-Language Models** *Under Review*
Mingxian Li*, Kun Xiang*, Xiao Dong*, Haoyuan Li, Zhao Zhen-Hua, Jingbin Tan, Terry Jingchen Zhang, Haiyang Hu, Yangle Fang, Kunkun Liu, Changzheng Zhang, Dandan Tu, Xiaodan Liang

*Equal contribution.

TECHNICAL SKILLS

Programming Languages: Python, C, C++.

Deep Learning Frameworks: PyTorch, TensorFlow.

Development & Computing Tools: Linux, Git/GitHub, Docker, Conda, Slurm, Jupyter Notebook, VS Code.

Writing & Presentation: LaTeX, Markdown, Microsoft Office, draw.io.

SCHOLARSHIPS

- Outstanding Student Scholarship (Third Prize), Sun Yat-sen University 2026
- Special Scholarship, Sun Yat-sen University 2025